

Discrete Mathematics Problem Sheet 1

Jan 16, 2026

1. Consider the power set of $\mathbb{N}$ as $\{0, 1\}^{\mathbb{N}}$. A partial order for it is defined as follows. given two elements $\mathbf{a}, \mathbf{b} \in \{0, 1\}^{\mathbb{N}}$, that is, sequences $\mathbf{a} = (a_1, a_2, \dots)$ and $\mathbf{b} = (b_1, b_2, \dots)$, define $\mathbf{a} \leq \mathbf{b}$ if $a_i \leq b_i$ for all $i \in \mathbb{N}$. An antichain is a subset of a partially ordered set such that any two distinct elements in the subset are incomparable.
 - (a) Give a countably infinite chain in $\{0, 1\}^{\mathbb{N}}$.
 - (b) Find a countably infinite antichain in $\{0, 1\}^{\mathbb{N}}$.
 - (c) Find an uncountable antichain in $\{0, 1\}^{\mathbb{N}}$.
 - (d) (*+) Find an uncountable chain in $\{0, 1\}^{\mathbb{N}}$.
 - (e) (*-) Find an uncountable collection of subsets of $\mathbb{N}$ such that any two subsets have finite intersection.
2. Are the sets below countable or uncountable? Justify.
 - (a) All non-increasing functions from $\mathbb{N}$ to $\mathbb{N}$. Non-increasing function f means $f(i) \geq f(i+1)$ for all $i \geq 1$.
 - (b) All non-decreasing functions from $\mathbb{N}$ to $\mathbb{N}$. That is, functions f such that $f(i) \leq f(i+1)$ for all i .
 - (c) All injective functions from $\mathbb{N}$ to $\mathbb{N}$.
 - (d) All surjections from $\mathbb{N}$ to $\mathbb{N}$.
 - (e) All bijections $\mathbb{N}$ to $\mathbb{N}$.
3. An infinite binary sequence is *lonely* if there are no consecutive 1's in it. Show that the set of lonely sequences is uncountable.
4. (The Hausdorff maximal principle). Let S be a partially ordered set. Then S contains a maximal chain (i.e. a chain which is not contained in a bigger chain). Use Zorn's lemma to prove this.
5. (Ultrafilter lemma) A *filter* on a set X is a set of subsets $\mathcal{F} \subseteq 2^X$ satisfying

- (a) $X \in \mathcal{F}$ (intuitively, the whole set is *large*).
- (b) $\emptyset \notin \mathcal{F}$ (the empty set is not large).
- (c) If $A \in \mathcal{F}$ and $B \in \mathcal{F}$ then $A \cap B \in \mathcal{F}$ (large sets have large intersection).
- (d) If $A \in \mathcal{F}$ and $B \supseteq A$ then $B \in \mathcal{F}$ (any set containing a large set is large).

A filter $\mathcal{F}$ on X is an *ultrafilter* if for any $A \subseteq X$, $A \in \mathcal{F}$ or $X \setminus A \in \mathcal{F}$ (every set is either large or co-large). Using Zorn's lemma, prove that every filter is contained in an ultrafilter.

6. Let X be an infinite set, and $\mathcal{F}$ a family of *finite subsets* of X . A subfamily $\mathcal{F}'$ of $\mathcal{F}$ is called a Δ -system if there is a fixed subset $S \subset X$ such that $A \cap B = S$ for all distinct $A, B \in \mathcal{F}'$.
 - Suppose all sets in the family $\mathcal{F}$ are of size n , for a natural number n . Show by induction on n that $\mathcal{F}$ contains an infinite Δ -system.
 - Suppose the family $\mathcal{F}$ is uncountable. Show that $\mathcal{F}$ contains an uncountable Δ -system.
7. Let κ be an arbitrary infinite cardinal number.
 - Can you define a vector space over any field F such that the set of vectors V has cardinality κ ?
 - Is there a group G of cardinality κ ? Can you make it abelian, nonabelian? How about fields or noncommutative rings of that cardinality?
 - Show that every group G of cardinality κ has a subgroup of cardinality $\aleph_0$.
 - Can you find similar substructures for rings and fields that you constructed above?